1. **<u>Course number and name</u>**
   CHE 201. Introduction to Thermodynamics

2. **<u>Credits, contact hours and categorization</u>**
   Credit hours: 3; Contact hours: 3 lecture hours per week; 50 minute optional discussion section per week; Engineering

3. **<u>Instructor's or course coordinator's name</u>**
   Betul Bilgin

4. **<u>Text book, title, author, publisher, year</u>**
   Fundamentals of Engineering Thermodynamics, Moran/Shapiro/Boettner/Bailey, 9th Edition (Zybooks version is used)

5. **<u>Specific course information</u>**
   a. <u>Brief description of the content of the course (catalog description)</u>
   Work and energy; conversion of energy; theory of gases and other states of matter; applications to energy conversion devices. Second law of thermodynamics, entropy, and equilibrium, with applications.
   b. <u>Prerequisites or co-requisites</u>
   Math 181, Calculus II and Phys 141, General Physics I (Mechanics).
   c. <u>Indicate whether a required, elective, or selected elective.</u>
   Required.

6. **<u>Specific goals for the course</u>**
   a. <u>Specific outcomes of instruction.</u>
   Students will learn to:
   - Apply energy balances for closed systems and control volumes.
   - Evaluate thermodynamic efficiencies of compressors, turbines, power plants, refrigeration and heat pump cycles.
   - Identify phases at a given state and find thermodynamic properties such as internal energy, specific volumes from steam tables.
   - Draw and interpret pressure-enthalpy (P-H), pressure-specific volume, temperature specific volume, temperature-entropy charts.
   - Relate thermodynamic properties via partial derivate, Maxwell's relations.
   - Interpret phase diagrams of binary systems.
   - Apply the entropy balance and identify if the process is reversible or irreversible.

Project

One, semester long project related to thermodynamics concept. Students will be working with groups and are expected to design a product or process with the focus of sustainability. Project includes research phase, modeling phase, prototype phase and testing phase. Design competition will be held among all the groups. All the groups are required to create Adobe Spark webpage for their design.

b. <u>Explicitly indicate which of the student outcomes are addressed by the course.</u>

Student Learning Outcome 1 Not Covered, 2 Introduced, 3 Reinforced, 4 Emphasized	1	2	3	4
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.		X		
(3) an ability to communicate effectively with a range of audiences.		X		
(4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.		X		
(5) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.		X		
(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.		X		

7. <u>Brief list of topics to be covered</u>

I. Introduction
II. Fundamentals (Chapter 1)
III. The First Law (Chapter 2)
IV. Properties of Pure Fluids (Chapter 3)
V. Control Volume Analysis (Chapter 7)
VI. The Second Law of Thermodynamics (Chapter 6)
VII. Entropy (Chapter 4)
VIII. Thermodynamic Relations (Chapter 11 Section 1 through 5)
IX. Power Cycles (Chapter 9; select sections)

AIChE Academy eLearning tutorials:

Students are expected to complete two AIChE Academy eLearning tutorials online and submit their certificates of completion to the instructor. They can choose two of the Level I Curriculum as listed below:

- An Introduction to Managing Process Safety Hazards
- Introduction to Process Safety
- Hazard Recognition
- Identifying & Minimizing Process Safety Hazards

- Lab Safety